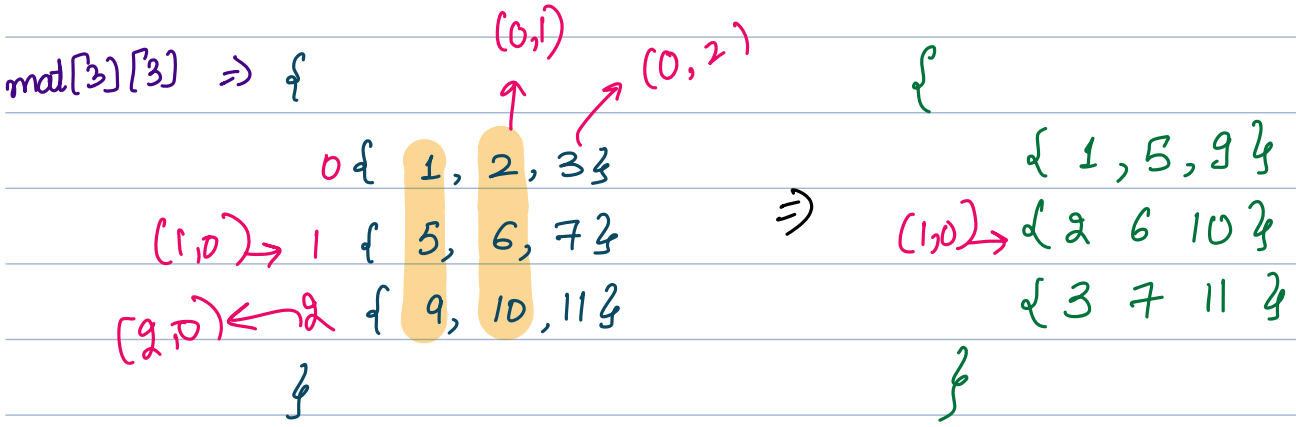
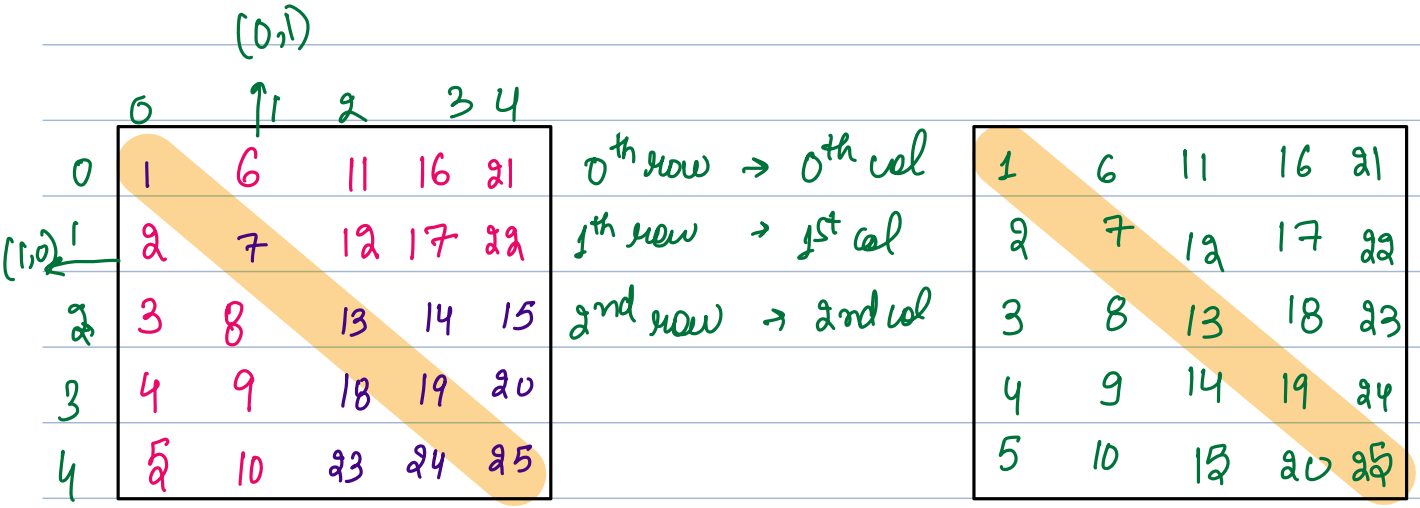
Good Evening Everyone :)

TRANSPOSE OF A SQUARE MATRIX

The transpose of a matrix is new matrix obtained by interchanging the rows & columns of the original matrix



Approach



TC: $O(N^2)$

SC: $O(1)$

```
function find_transpose (matrix[N][N]) {
    for (i → 0 to N-1) {
        for (j → i+1 to N-1) {
            swap (matrix[i][j], matrix[j][i])
        }
    }
}
```

ROTATE A MATRIX TO 90 DEGREE CLOCKWISE

1	2	3
4	5	6
7	8	9

7	4	1
8	5	2
9	6	3

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

1	6	11	16	21	→ reverse
2	7	12	17	22	→ reverse
3	8	13	18	23	→ reverse
4	9	14	19	24	→ reverse
5	10	15	20	25	→ reverse

21	16	11	6	1
22	17	12	7	2
23	18	13	8	3
24	19	14	9	4
25	20	15	10	5

Calculate transpose + Reverse Every Row = Rotate by 90°

Approach:

TC : $O(N^2)$
SC : $O(1)$

```
function rotate (mat[N][N]) {
    mat = transpose (mat)
    for (i -> 0 to N-1) {
        reverse (mat[i])
    }
    return mat
}
```

FIND THE MAXIMUM NUMBER OF CONSECUTIVE 1's AFTER REPLACEMENT

Given an array of 1s and 0s, you are allowed to replace only one 0 with 1. Find the maximum number of consecutive 1s that can be obtained after making replacement

Input :- [1 1 0 1 1 0 1 1]

5

5

Ans = 5.

Quiz2.

A[] = [1 1 0 1 1 0 1 1 1]

⇒ 6.

[0 1 1 1 1 1 1.]

total Ones ⇒ 6

0 4

1 1 1 1 0 0

3N

⇒ O(N).

2N

~~#6~~ N

Quiz 3.

A[] = [0 1 1 1 0 1 1 0 1 1 0]

[0 0 1 1 0]

Code :-

```
function findMaxConsecutiveOnes (nums[]) {

    n = nums.length;
    maxCount = 0
    totalOnes = 0

    for (i -> 0 to n-1)
        if (nums[i] == 1)
            totalOnes ++
        }
    }

    if (totalOnes == n)
        return n
    }
```

```
for (i -> 0 to n-1) {
    if (nums[i] == 0) {

        int l = 0 , r = 0 , j = i+1

        // find consecutive 1's at right
        while (j<n and nums[j] == 1)
            r++
            j++
        }

        // find consecutive 1's at left
        j = i-1

        while (j >= 0 and nums[j] == 1)
            l++
            j--
        }

        count = l + r+1
        if (count > maxCount)
            maxCount = count
    }
    print (maxCount)
}
```

CONSECUTIVE ONES VARIATION

given an array of 1s and 0s, you are allowed to replace only one 0 with 1. Find the maximum number of consecutive 1s that can be obtained by SWAPPING at most one 0 with 1 (already present in string)

⇒ 4

Input: [1 0 1 1 0 1] ⇒ 4.

4

ans = 3/5

Total ⇒ 6.

[1 1 0 0 1 1 0 1 1 0]

l = 2
r = 2

l + r + 1
2 + 2 + 1

Quiz.

[1 1 0 1 1 1]

⇒ 5.

Code :

TC : O(N)

SC : O(1)

```
function findMaxConsecutive(nums[]) {
    n = nums.size
    maxcount = 0
    totalOnes = 0

    for (i -> 0 to n-1)
        if (nums[i] == 1)
            totalOnes ++

    if (totalOnes == n)
        return n

    for (i -> 0 to n-1) {
        if (nums[i] == 0){
            l=0, r=0, j=i+1
            while (j<n && nums[j]==1){
                r++
                j++
            }
        }
    }
```

```
        j = i-1
        while (j>=0 && nums[j]==1){
            l++
            j--
        }
        if (l+r == totalOnes)
            currmax = l+r
        else{
            currmax = l+r+1
        }
        maxCount = max(currmax, maxcount)
    }
}
```

REVERSE STRING WORD BY WORD

Given an input string s, reverse the order of words. A word is defined as sequence of non-space characters. The words in "s" are separated by atleast one space.

Return a string of words in reverse order, concatenated by single space. The returned string should have single space separating the words.

String s = "My Name is Twinkle"

"elkniwT si emaN yM"

"Twinkle is Name My"

s = "Hello World"

World Hello

"Twinkle"

// Trim the Spaces

↳ Spaces at left
↳ Spaces from right
↳ Ignore inbetween spaces.

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

" Scaler is best "

// Reverse the entire String

// Reverse each word in the string.